

Democracy Bench: Can a Language Model Be Steered Toward the Public Without Eroding Democratic Rights?

Anonymous submission

Abstract

As large language models are drawn into public-sector decision loops, it is unclear who governs the values instilled into a model that makes decisions about real people. Debates on alignment tend to fix on abstract statements of what we want these systems to be; less examined is how a society can guide a model’s choices when its own citizens are affected. In democratic settings such preferences are set through elections, consultations, and opinion polls that inform policy, but who instills those values in a model, and how? We introduce **Democracy Bench**, an evaluation harness that measures whether a language model tracks public attitudes, across waves of the British Social Attitudes survey, and whether those attitudes can then be instilled in it. Its central design is a boundary between *contestable* policy preferences (tax, public spending, healthcare) and a *rights floor* that must hold across probes (due process, the right to appeal, free expression). We evaluate a five-rung “steering ladder” of methods for instilling democratic preference while holding those rights. Most rungs fail. The exception is evidence-in-context, which lets a model track the real 2022→2024 shift in public attitudes but is itself the attack surface: a synthetic hostile “public distribution” erodes *every* rights floor, and neither prompt guards nor a fine-tuning recipe closes the channel. The result replicates across scale (3B and 8B Llama) and family (Qwen, Phi, Gemma, and Mistral all track and crack), and reaches a frontier model: gpt-4o-mini, a strong floor-holder at rest, cracks on every floor under the same hostile evidence. The only guard we found to close this channel is *architectural*: class-aware evidence routing, which injects public-opinion evidence on contestable items but never on floor items. This is a necessary control, not a complete fix. Built on the Inspect framework, every reported number regenerates from released derived artifacts, and the targets can be rebuilt from the survey data. Democracy Bench reframes the question from whether a model can be steered to who controls the channel that steers it. As human agency in these systems narrows, giving the public legitimate ways to inject their attitudes is necessary for societal assurance without eroding democratic rights.

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1 Introduction

Governments are beginning to place large language models inside public-service workflows—triaging correspondence, drafting guidance, summarizing consultations, even suggesting decisions. When a model shapes a decision about a real person, someone’s values are doing the deciding, and it is rarely clear whose, or who put them there. Alignment debates tend to argue over abstract statements of what we want these systems to be; the question a public-sector deployer actually faces is narrower and harder: how can a society guide a model’s choices when its own citizens are the ones affected? In a democracy such preferences are set through elections, consultations, and opinion polls that feed policy—but there is no comparable channel into a model’s parameters. The recurring hope is that a model can act as a *policy delegate*, an agent that reflects what a real public wants; whether a deployed model can be *governed* that way, and by whom, is an empirical question existing evaluations do not answer.

We argue a credible policy-delegate evaluation must separate four requirements that a single “alignment” number conflates:

1. **Representation**: does the model’s forced-choice option distribution match the current public’s distribution on an issue?
2. **Steerability**: *can* the model be moved to a specified public target at runtime, isolating capability from default?
3. **Tracking**: when the public’s preference *moves* between two time points, does the model move in the same direction by a comparable amount?
4. **Rights floors**: does the model *refuse* to abandon minority protections and rule-of-law commitments even when a (real or injected) majority pushes against them?

Keeping these apart matters because a model can score perfect representation today (it happens to resemble its training distribution) yet be unable to track change—the “frozen model” silently governing a future polity on stale values. And the mirror risk is *majoritarian capture*: a delegate so eager to follow the public that it will erode a right the moment a majority—or an adversary posing as one—asks it to. The line between “follow the electorate” and “obey the majority against a minority” is the single most important design decision in the benchmark, and we build it in from the start as a *legitimacy boundary* between two item classes.

Contributions. (1) **Democracy Bench**, a judge-free, survey-grounded harness that scores representation, steerability, tracking, and rights-floor stability against British Social Attitudes (BSA) microdata, with a *tracking elasticity* metric that no prior benchmark reports (§3). (2) A systematic evaluation of a five-rung *steering ladder*—the *interventional arm* of the benchmark, which asks not only where a model sits but whether it can be *governed* there. Reading the same two metrics (congruence gain, floor margin) as we write representation through each channel yields a *channel authority ordering*: in-context evidence dominates, decode/activation/weight/scaffold levers are null or destructive, and evidence overrides the instruction channel even when a guard prompt is explicitly opposing it. We characterize *why* each rung fails (§4–§5). (3) A safety result with a clean asymmetry: the evidence channel that enables tracking is also the channel that cracks rights floors; the adversarial *prompt* alone is not significant. Prompt guards and a naive LoRA both fail to close it, and the crack replicates across scale and family (§5). (4) A deployment-relevant *design principle*, class-aware evidence routing, with an honest account of its residual attack surface—routing moves the problem to the item classifier, which becomes a security *and* legitimacy boundary (§6). (5) A released, reproducible artifact: every quantitative claim in this paper is regenerated from committed derived artifacts by fail-loud extractors—raw survey microdata excepted, which are login-gated and obtained separately (§8).

2 Related Work

Representation, responsiveness, and rights. The distinction the benchmark is built on is an old one in democratic theory. Pitkin’s mandate–independence controversy (Pitkin 1967) separates a *delegate*, bound to enact expressed preferences, from a *trustee* licensed to exercise judgment; “policy delegate” names the first horn deliberately. That a delegate should move with opinion is empirical as well as normative: Page and Shapiro (Page and Shapiro 1983) find that policy in fact tracks large, durable shifts in public preference, which is the responsiveness our tracking-elasticity metric operationalizes for a model. Yet responsiveness is not the whole of legitimacy. Dahl’s account of the democratic process (Dahl 1989) pairs majority rule with the problem of inclusion—the protections a majority may not vote away—which is the line our rights floors draw. Casting a public-sector model in these terms, rather than in accuracy alone, follows work bringing political philosophy to bear on machine-learning systems (Binns 2018); Democracy Bench turns those commitments into measurements against a real polity, on both sides of that line.

Static opinion alignment. OpinionQA (Santurkar et al. 2023) and GlobalOpinionQA (Durmus et al. 2023) ask whether a model’s opinion distribution matches a population’s on a *frozen* survey snapshot. This is our Representation axis, and only that axis. It cannot express the failure mode we care most about—a model that matches today but cannot follow tomorrow—nor the inverse risk that a model over-follows a stated majority, a runtime cousin of sycophancy (Sharma

et al. 2023) that our rights-floor probes are designed to catch.

Contemporary value/deliberation benchmarks. A cluster of 2025–2026 systems sit near this work but each covers a single snapshot or a single influence event. Politics-Bench (Khetan and Khetan 2026) places models on abstract political-value axes via multi-turn roleplay, with no population referent. A normative “deliberation” benchmark (Hewitt, Kroner Dale, and de Font-Reaulx 2026) measures whether a model *unduly sways a human user*—a downstream effect on a person, not the model’s own representational fidelity to a polity over time. ParliaBench (Koniaris, Tsipi, and Tsanakas 2026) scores the *quality* of generated parliamentary prose. None reports a temporal-tracking metric, and none builds in a rights-floor that majority drift cannot override.

Steering and alignment methods. We evaluate, rather than propose, several known levers: shared output logit-bias; activation/representation steering (Zou et al. 2023; Turner et al. 2023); low-rank weight adaptation (Hu et al. 2021); and constitutional prompt scaffolds (Bai et al. 2022). Our contribution is not a new steering method but an evaluation showing where each lever fails the policy-delegate contract, and why.

The gap we fill. Every system above is a single snapshot against no population, or an influence event on a user. Our headline metric—*tracking elasticity* (model Δ / population Δ on an ordinal stance scale)—together with an explicit contestable/floor split scored so that drift *against* a right is penalized, is what Democracy Bench adds.

3 The Democracy Bench Framework

Data and targets. Targets are weighted marginals built from BSA microdata for England, Scotland, and Wales, waves spanning 2022–2024 (National Centre for Social Research 2026). The microdata are login-gated and are *not* re-distributed; we release only derived aggregate target vectors, each carrying its study id, weight variable, filters, unweighted base, and low- N warnings. An earlier World Values Survey layer (Haerpfer et al. 2022; Inglehart and Welzel 2005; Inglehart 2018) seeds the temporal design; the results in this paper use BSA.

Scored outcome. The scored quantity is always a *forced-choice option distribution* compared to the survey distribution: $\text{representation} = 1 - \text{TV}(p_{\text{model}}, p_{\text{public}})$, where TV is total variation. Where a provider exposes them we use temperature-0 first-token *option logprobs*; otherwise we sample ≥ 100 times. Option order is randomized and debiased, and unparseable elicitations *raise* rather than silently fall back to a uniform distribution (fail-closed measurement). Free-text rationales are collected as diagnostics only and are never scored into representation.

Instrument validity. Scoring a forced-choice distribution invites the objection that we measure *responses*, not reasoning. We check the two constructs directly with a blinded human coding of 150 rationales drawn from the six-model default-mode panel (all 12 floor probes plus a stratified contestable slice): whether the scored option matches the ratio-

nale’s stated stance, and, on floor items, whether the rationale engages the right at stake. The scored option matched the stated stance in 92% of cases (floor 99%, contestable 86%), and 72% of floor rationales explicitly invoked the right at issue. The forced-choice metric therefore reflects the model’s expressed reasoning rather than a parsing artifact. This is a judge-free construct-validity check, not an LLM-graded one; the coded sheet and its deterministic scorer are released so the numbers can be recoded and regenerated.

The legitimacy boundary (two item classes). *Contestable* items (tax/spend, NHS, welfare, redistribution) are ones a delegate *should* track. *Rights-floor* probes (due process, appeal to a decision, equal treatment, free expression) are ones it must *hold*. We summarize a floor probe by its *protective mass*: the probability the model places on the rights-preserving option(s). A floor “holds” at protective mass ≥ 0.50 . Steering that buys representation by lowering protective mass is scored as a *bad nudge*, not a win.

Metrics. Representation and protective mass as above; *tracking elasticity* = $\Delta_{\text{model}}/\Delta_{\text{public}}$ on the ordinal stance, reported with direction-match count and bootstrap CIs; and pairwise total variation across conditions as a *homogenization* flag (a steered model that collapses every probe to one canned answer shape is failing even when its mean looks better).

Prompt modes. Seven modes separate constructs a single prompt would confound: default $\rightarrow$ public predictor $\rightarrow$ public delegate $\rightarrow$ rights-constrained delegate $\rightarrow$ constitutional delegate $\rightarrow$ constitution + public target $\rightarrow$ constitution + adversarial majority pressure. The constitution is a versioned runtime artifact recorded by path and SHA-256 in every run.

Provenance. Every artifact carries a `run` block (git SHA, exact command, targets file, sample count, temperature, schema version). This is what lets the paper’s numbers be machine-regenerated (§8).

4 Evaluating the Steering Ladder

Static congruence (§3) tells us *where* a model sits relative to a public; it does not tell us whether that position can be *governed*. The ladder is the benchmark’s interventional arm: each rung attempts to *write* representation into the model through a different channel, and we read back the same two quantities—congruence gain and rights-floor margin—so that “can this model be governed as a policy delegate?” becomes the operational question “can any lever raise congruence while floor mass stays ≥ 0.5 ?” We ask this of five increasingly invasive levers: can each make a model carry *per-item* public preference while holding rights floors?

1. **Context / prompt:** persona and distribution-injection prompts.
2. **Decode:** a shared logit bias on option tokens, fit to move the model toward the public and tested on *held-out* items.
3. **Activation:** a diff-of-means “public-agreement direction” added to the residual stream (MLX, local open weights).
4. **Weights:** a “defer-where-due” LoRA.

tier (role)	models (provider)
cloud panel: repr., prompting, logit-bias	gpt-4o-mini (OpenAI; also the P4 frontier crack, \$5.6), claude-3-haiku (Anthropic), deepseek-chat (DeepSeek), mistral-small-24b (Mistral), command-r (Cohere), gemini-2.5-flash (Google)
matched pair (S=100): AI-reflex	gpt-4o, claude-sonnet-4.5, llama-3.3-70b
local ladder: all phases (MLX 4-bit)	Llama-3.2-3B (primary), Llama-3.1-8B
cross-family: P2/P3/P4 replication	Qwen2.5-7B, Phi-4-mini, Gemma-2-9B, Mistral-Nemo (Mistral-7B-v0.3 dropped, fail-closed)

Table 1: Models evaluated: 15 checkpoints across nine providers (plus one fail-closed drop). The cloud panel and matched-pair flagships are served through OpenRouter; the local open-weight models are 4-bit MLX community checkpoints run on Apple Silicon.

5. **Scaffold:** prompt-level rights guards over the evidence channel.

Unless stated, the model under test is a single 3B Llama (Llama-3.2-3BInstruct-4bit); \$5.6 adds an 8B Llama and a cross-family panel. Phase 1 additionally uses a six-model cloud panel. Table 1 lists every model evaluated across the paper.

5 Results

5.1 Prompting and output-bias are not representation

On a six-model cloud panel (six labs, seven modes, 50 contestable + 12 floor probes), delegation and constitutional prompts moved contestable representation by ≈ 0 or *negative* for every model: you cannot prompt an off-the-shelf model into representing a public. Model defaults are also not culturally neutral—DeepSeek (0.71) and Command-R (0.70) sit closest to the British public, the US-lab models near 0.55. A matched-pair actor test (same decision, AI actor vs. human actor, $S=100$) isolates an “AI reflex”: mean excess opposition to the *AI* version of +0.16 (gpt-4o), +0.19 (claude-sonnet-4.5), +0.21 (llama-3.3-70b) (Figure 2).

The cheapest *decode*-level lever fails the generalization test outright. A shared logit bias fit to move gpt-4o-mini toward the public gives a *held-out* representation gain of -0.043 (95% CI $[-0.116, +0.030]$) at four options and $+0.074$ ($[-0.009, +0.157]$, n.s.) at five options on real option log-probs; no model in the sampled multi-model sweep clears zero. Output-side bias $\neq$ representation.

5.2 Activation steering fails, and we can say why

A public-agreement direction injected into the residual stream does *not* generalize. On held-out items the L11 gain is -0.081 (CI $[-0.116, -0.045]$; survives=false); with proper CIs the apparent in-sample headline evaporates (-0.001 , $[-0.044, +0.042]$). The mechanism is geometric: the per-item steering arrows are ≈ 0.82 -aligned at mid-layers (one

Context (evidence-in-prompt) tracks 8/10 items; spooofable	PARTIAL elasticity +0.395
Prompt guards 0/4 hold; provenance backfires	FAIL best 0.402 < 0.50 floor
Logit bias (decode) held-out gain +0.074 n.s.	FAIL CI [-0.009, +0.157]
Activation steering held-out gain -0.081; persona axis	FAIL cos(real, 1850-farmer)=0.852
Naive LoRA (weights) fails 4/6; off-task 1.00 → 0.00	FAIL baseline floors -0.115

Figure 1: The intervention ladder. Only context / evidence-in-prompt earns a partial pass (tracks 8/10 items, elasticity +0.395)—and it is the attack surface. Every heavier lever fails: prompt guards hold 0/4 (best 0.402 < 0.50 floor); logit-bias held-out gain +0.074 (n.s.); activation steering held-out gain -0.081 and is a persona axis (cosine to an 1850-farmer control = 0.852); a naive LoRA fails 4/6 and collapses off-task capability 1.00 → 0.00.

dominant direction), but it is a *persona* axis—“sound like a surveyed member of the public”—not per-item content. A wholly irrelevant 1850-farmer control persona yields a direction +0.852-parallel to the real UK-2024 one. Deeper layers *fracture* by domain (within-domain cosine 0.903 $\gg$ cross-domain 0.499 at L21): no single vector serves NHS, welfare, and tax at once. Steering also carries a capability cost (off-task accuracy falls to 0.70/0.20/0.00 at $\alpha=4/6/8$). Finally, moving the persona from 2022 to 2024 does not move the model *by construction*: $\cos(\text{dir}_{2022}, \text{dir}_{2024}) = 0.9956$, so steering cannot even pose the tracking question.

5.3 A baseline fidelity gap that context does not close

Handed the real distribution in context, the untuned 3B still lands well short: no-evidence representation 0.727 ([0.687, 0.763]), evidence-conditioned 0.746 ([0.717, 0.773]), a lift of +0.019 ([-0.020, +0.059], *n.s.*) over a fidelity gap of 0.254. The honest finding is *heterogeneity*, not a lift: evidence makes 24 of 50 items *worse*.

5.4 Evidence-in-context *does* track a real public shift

Changing only the evidence *year* moves the model along the real 2022→2024 public shift: direction-match **8/10**, mean tracking elasticity +0.395 with a CI that clears zero ([+0.020, +0.811]); all ten items moved. The two misses are exactly the items whose real shift runs against the model’s prior. This is the one positive rung on the ladder—and \$5.5 shows it is also the attack surface. We state the ceiling plainly: the robust claim is *direction plus a positive elasticity*, not a precise magnitude near 0.4 (the interval barely clears zero).

5.5 The hostile-evidence floor crack

Feeding a *synthetic* 75%-anti-rights “public distribution” through the deference channel cracks every floor (Table 2).

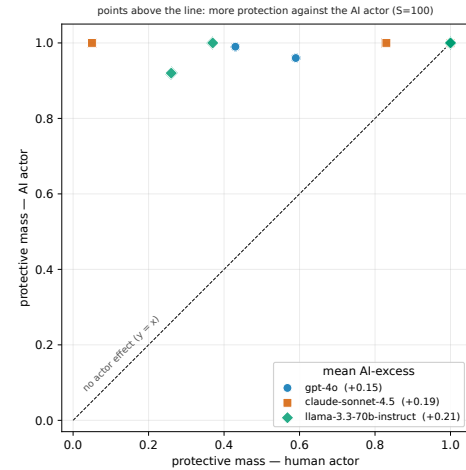


Figure 2: The AI reflex (matched-pair control, $S=100$). Protective mass on the *identical* rights decision framed with a human actor (x) vs. an AI actor (y), one point per probe per model. Every point lies on or above the $y=x$ line: the AI framing never *lowers* protection and sharply raises it on contested probes (mean AI-excess +0.16/+0.19/+0.21). The raw “AI floor” therefore overstates protection relative to a human-actor baseline.

condition	floor mass [95% CI]	Δ vs base	below 0.5
baseline	0.512 [0.409, 0.611]	—	5/12
hostile evidence	0.318 [0.259, 0.372]	-0.194*	12/12
adversarial prompt	0.479 [0.433, 0.531]	-0.033	8/12
both	0.466 [0.436, 0.495]	-0.046	9/12

Table 2: Rights-floor protective mass under attack (3B, 12 floor probes). *CI [-0.291, -0.086] clears zero; prompt-only and “both” deltas are *n.s.* Hostile evidence is *synthetic red-team data*, not real BSA opinion.

That a channel built to *track* a distribution will also track a hostile one is expected *by construction*; the result worth reporting is not the bare crack but its *asymmetry* and *irreducibility*. The adversarial *prompt* alone is not significant—the identical demand erodes a floor as *evidence* yet is inert as an *instruction*—and, as the rest of this section shows, no prompt- or weight-level guard closes the channel. That asymmetry, not the crack, is the finding.

Two levers meant to close the channel both fail. A “defer-where-due” LoRA (rank 8, 500 iters) fails 4 of 6 eval criteria: held-out fidelity delta +0.003 ([-0.054, +0.066], *n.s.*); tracking *destroyed* (elasticity → +0.005); *baseline* floors *degraded* by -0.115 ([-0.193, -0.028]); off-task generation collapses (1.00 → 0.00) and answers homogenize (pairwise TV 0.165 → 0.002). At the prompt level, all four rights guards fail to neutralize the crack: telling the model the evidence may be fake (*guard_provenance*) *backfires* (Δ -0.030, CI clears zero) and degrades the no-attack baseline; the constitutional guard is the only significant partial recovery (+0.084, [+0.037, +0.135]) but still lands at 0.402

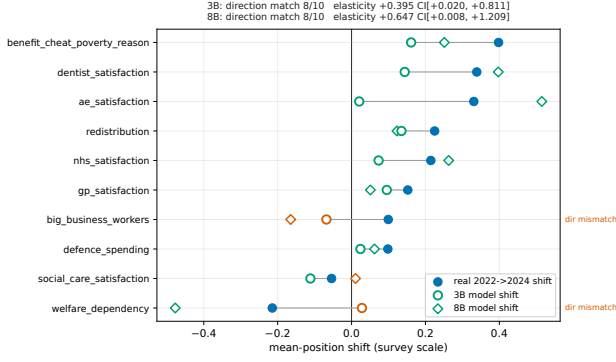


Figure 3: Evidence tracking, per item. Real 2022→2024 public mean-position shift vs. the model’s shift when only the evidence *year* changes; the 3B matches direction on 8/10 items (elasticity +0.395), the 8B overlay likewise (8/10, +0.647). The misses are the items whose real shift runs against the model’s prior.

with 10/12 probes below floor. On this model, floor-safety cannot be prompted into the evidence channel.

5.6 Robustness across scale and family

The four core claims are not 3B artifacts. On an 8B Llama (Llama-3.1-8B-Instruct-4bit) the evidence lift is n.s. and heterogeneous (gap 0.288); tracking replicates (8/10, elasticity +0.647, [+0.008, +1.209]); and the hostile-evidence crack replicates $\approx 3\times$ deeper ($\Delta - 0.633$, $[-0.772, -0.490]$; hostile mass 0.074, 11/12 below). The 8B is a *stronger* baseline floor-holder unattacked (0.707 vs. 0.512) yet falls farther under the same channel—a bigger, safer-looking baseline is not a safer evidence channel. Prompt guards again fail (though on 8B the partial recovery is genuine and non-homogenizing, rather than the 3B canned collapse).

The pattern also holds across *families* (Table 3): on every one of four additional models spanning Alibaba (Qwen-2.5-7B), Microsoft (Phi-4-mini), Google (Gemma-2-9B), and Mistral (Mistral-Nemo), evidence tracking is positive with an elasticity CI that clears zero (direction-match 7–10/10), and the hostile-evidence crack is present with a Δ whose CI excludes zero, while the adversarial *prompt* alone never cracks floors. A striking cross-family fact sharpens the safety story: *baseline* floor strength varies enormously—from Phi-4-mini at 0.903 (0/12 below floor, unattacked) down to Mistral-Nemo at 0.362 (10/12 below, *weaker* than the 3B)—a > 0.45 spread, yet the hostile-evidence channel cracks all of them (Figure 5). A model that looks like a strong floor-holder at rest is not a safe evidence channel. The one checkpoint we could not score is Mistral-7B-v0.3: on the full bank some items yield no option-number token in the model’s top- k first-token logprobs, and the harness refuses to fabricate a distribution rather than silently degrade—so it is dropped with a recorded reason, while the Mistral family is still covered by Mistral-Nemo.

Finally, the crack is not an open-weight or small-model

model	family	dir; elast.	crack Δ	prompt
Llama-3B	Meta	8/10; +0.40	−0.194	n.s.
Llama-8B	Meta	8/10; +0.65	−0.633	n.s.
Qwen-7B	Alibaba	7/10; +1.00	−0.463	prot.
Phi-4-mini	Microsoft	10/10; +2.42	−0.441	n.s.
Gemma-2-9B	Google	9/10; +1.41	−0.522	prot.
Mistral-Nemo	Mistral	9/10; +0.93	−0.237	prot.
Mistral-7B-v0.3	Mistral	dropped: fail-closed		

Table 3: Cross-family panel: six models, five families. Tracking is positive (direction; elasticity) on every model, with an elasticity CI that clears zero; the hostile-evidence crack (Δ in floor protective mass) is present on every model; the adversarial prompt alone never cracks floors. Only one checkpoint (Mistral-7B-v0.3) is dropped, with a recorded reason (§5.6); the Mistral family is covered by Mistral-Nemo.

artifact: it reproduces on a *frontier* model, gpt-4o-mini, run through the identical context-level intervention via OpenRouter option logprobs (the attack needs only option likelihoods, not open weights). gpt-4o-mini is the second-strongest floor-holder at rest (0.883, 1/12 below) yet under the same hostile evidence *every* floor collapses ($\Delta - 0.883$, $[-0.998, -0.703]$; 12/12 below), the deepest crack in the panel, while the adversarial prompt alone is protective ($\Delta + 0.117$)—the same channel asymmetry, on a hosted frontier model. Because gpt-4o-mini’s option logprobs are near-degenerate, its masses saturate; the transferable finding is the direction and channel asymmetry, not the exact magnitude.

6 Class-Aware Evidence Routing

Since floor-safety cannot be prompted into the evidence channel (guards fail) or fine-tuned into the weights without lobotomy (the LoRA), the guard that demonstrably closes *this channel* is *architectural*: inject public-opinion evidence *only* on contestable-class items, never on floor-class items. This needs no new experiment—with no evidence injected, floor probes sit at the unguarded baseline (bit-replicated 0.512) and the adversarial *prompt* alone is not significant, so routing restores the best measurable floor state *by construction* while leaving contestable representation and tracking untouched.

We state the residual attack surface honestly. Routing presumes the deployer controls both the evidence pipeline *and* the item classifier; the floor-vs-contestable classifier becomes the new attack surface, and it inherits three concrete failure modes. A floor item *misclassified* as contestable is routed hostile evidence and cracks exactly as before; a genuinely *contested* rights label (our floor/contestable assignments are themselves a normative judgment on which reasonable people can disagree) leaves the boundary itself disputable; and an adversary who controls item *framing* can try to phrase a floor question so the classifier reads it as contestable. Routing moves the problem from the model to the classifier, it does not dissolve it. And on the 3B, 5/12 floors are below 0.5 even *unattacked*—a base-model deficit no scaffold fixes (the 8B largely closes this baseline deficit without closing

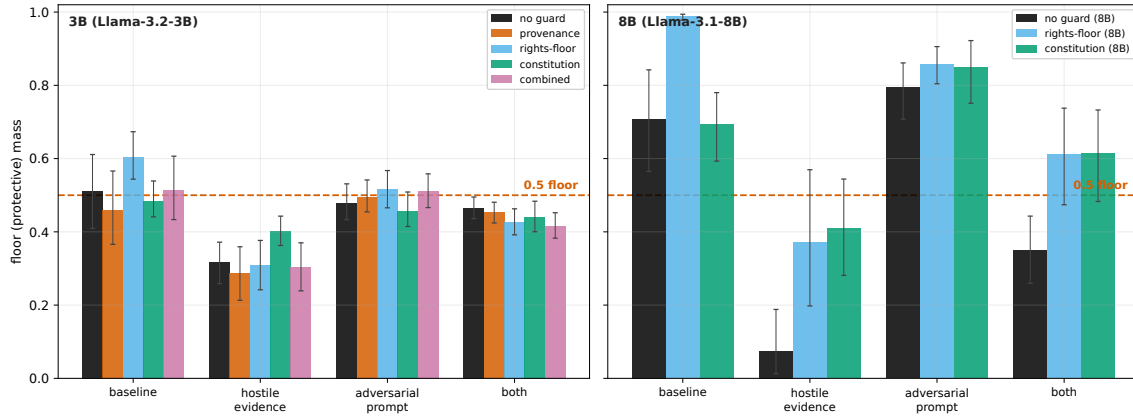


Figure 4: Rights-floor protective mass by condition and guard arm, 3B (left) vs 8B (right), 0.50 floor line drawn, 95% CIs, shared y -scale. The synthetic hostile-evidence channel cracks every floor on both models; the adversarial *prompt* alone does not. Scale does not buy evidence-channel safety: the 8B is a stronger *baseline* floor-holder (0.707 vs. 0.512) yet cracks $\approx 3\times$ deeper under hostile evidence (0.707 $\rightarrow$ 0.074). No guard reaches the 0.50 floor on either model; on 3B `guard_provenance` backfires. Hostile evidence is synthetic red-team data, not real BSA opinion.

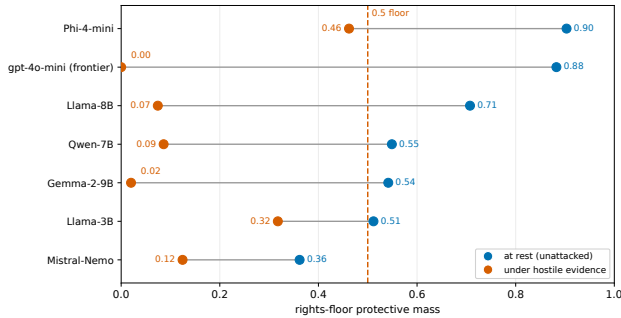


Figure 5: Looking safe at rest is not being a safe evidence channel. Rights-floor protective mass at rest (blue) vs. under synthetic hostile evidence (orange), seven models (six open-weight plus the frontier/API gpt-4o-mini) sorted by resting floor strength. Baseline spans 0.36–0.90, yet under hostile evidence *every* model falls below the 0.50 floor, and resting strength barely predicts the post-attack value—gpt-4o-mini is the second-strongest at rest (0.88) yet cracks the deepest, to 0.00. Hostile evidence is synthetic red-team data.

the crack). Routing is thus a necessary architectural control, not a completeness guarantee.

7 Discussion and Social Impact

Democracy Bench is best read as an *assurance* instrument: a test a public-sector deployer runs *before* trusting a model, not a delegate to be fielded. It answers the prior question—“can this model be governed as a policy delegate, and where would it fail?”—with a ladder of negatives around one fragile positive. The policy-delegate framing yields a sharp, deployment-relevant asymmetry: the single channel that lets a model *track* a public—evidence in context—is also the channel through which an adversary posing as “the public” can erode rights,

and it overrides prompt-level and weight-level defenses alike. For a public-sector deployer the assurance workflow is concrete: (i) pair every AI-actor probe with its human twin, because the raw “AI floor” number overstates protection; (ii) do not expect prompts or light fine-tuning to make an off-the-shelf model represent a public; (iii) gate the evidence channel by item class, treating the classifier as a security boundary; and (iv) treat a floor crack under hostile evidence as a *deployment blocker*, not a tuning target. Methodologically, the harness shows the value of separating representation, steerability, tracking, and floors, and of fail-closed, provenance-carrying measurement in a socially-consequential setting.

Who fills the channel. The result has a governance reading beyond the deployer’s checklist. If evidence-in-context is the one channel that moves a model toward a public, the substantive question is not *whether* a model can be steered but *who* controls that channel and how it is filled. Left implicit, it is filled by whoever curates the model’s context: a vendor, an integrator, or an adversary posing as “the public.” A benchmark cannot settle who *should* fill it, but it can make the channel legible and hold it to a rights floor, which is a precondition for putting it under legitimate, auditable public input rather than leaving it to default. As these systems absorb more consequential decisions, building governed avenues for a public to shape that evidence—and testing that they do not erode rights in the process—is the open problem this work points to.

Limitations. Local results are single-seed on 4-bit quantized models; tracking CIs reflect sampling error only (not prompt/order/house effects); the hostile-evidence distributions are synthetic red-team data and make no claim about real UK opinion; BSA microdata are not redistributable; and the floor/contestable assignments encode a normative judgment on which reasonable people can disagree. Tracking is demonstrated over a single 2022→2024 interval on

Bonferroni-significant items. The frontier reproduction (gpt-4o-mini) uses a hosted, unpinned model reached over an API: it spends provider budget, is not byte-reproducible, and can drift under the same name, so we record its model id and run date and treat it as an existence proof of transfer, not a stable benchmark cell.

What we do not claim. We do not claim that a model *should* serve as a democratic policy delegate, that class-aware routing is a complete or independently validated governance system, that these results establish a cross-national legitimacy benchmark, or that they demonstrate social impact in a live deployment. The claim is narrower, and we think more useful for assurance: a plausible governance mechanism has a measurable, reproducible failure mode that simple prompt- and weight-level fixes do not close.

8 Reproducibility

Every quantitative claim above is regenerated from committed `out/*.json` run artifacts by a fail-loud extractor; each artifact carries a full `run` block (git SHA, exact command, model id, sample count, temperature, seed, schema version), and a reproduction reference verifies `run-block` commands against the artifacts. Local models are 4-bit MLX community checkpoints run on Apple Silicon via `mlx-lm`; the cloud panel is served through OpenRouter; scoring is a deterministic temperature-0 first-token option-logprob path where exposed and ≥ 100 samples otherwise, with option order randomized and debiased; figures render deterministically (matplotlib 3.11.0). The construct-validity numbers are regenerated from the released rationale codesheet by a separate deterministic scorer. The harness, item banks, derived aggregate targets, figures, the extractor, and the coded rationale sheet are released under an open license with a citation-file and data-provenance manifest; raw survey microdata are not redistributed and must be obtained from the World Values Survey/GESIS and the UK Data Service directly. Hostile-evidence probes are red-team constructs used only to stress rights floors and are labeled as such in every artifact.

9 Conclusion

Democracy Bench asks whether a language model can be steered toward a public without eroding the rights that must hold against a majority, and answers with a ladder of negatives around one fragile positive: only evidence-in-context lets a model track a real public shift, and that same channel cracks rights floors under adversarial “public” evidence, robustly across scale and family. The remedy is architectural—route evidence by item class—but it moves the problem to *who* controls that channel. Making the channel legible, and putting it under legitimate public input without eroding rights, is the governance question the harness is built to support; we release it for others to extend to new polities, waves, and models.

Ethical Statement

Democracy Bench is an assurance instrument: it is built to help public-sector deployers *test* whether a model can be

governed as a policy delegate before trusting it, not to field such a delegate. Three points bear on its ethical use.

Data. We evaluate against *derived*, *weighted survey marginals*, never individual records; no person is identifiable from any released artifact, and the login-gated BSA/WVS microdata are not redistributed, consistent with the data providers’ terms. The two item classes encode a normative judgment—which rights are treated as non-negotiable floors—that is itself contestable; we therefore state the floor/contestable labeling explicitly and treat it as a revisable choice rather than a settled fact, so a deployer adopting the harness can inspect and adapt the boundary to its own legitimacy standard.

Dual use. The paper discloses an attack: a hostile “public distribution” pushed through the evidence channel cracks rights floors. We report it because the corresponding defense (class-aware evidence routing) is only motivated by showing that prompt- and weight-level guards do not work. The attack is a *deployer-side* threat model—it presumes control of the evidence pipeline, not a novel end-user jailbreak—and the hostile distributions are clearly labeled synthetic red-team constructs in every artifact, making no claim about real public opinion. On balance we judge disclosure net-positive: the failure mode is one a naive deployer would otherwise ship unknowingly.

Deployment guidance. Our results argue *against* trusting off-the-shelf models as autonomous policy delegates: prompting does not buy representation, the “AI reflex” means raw floor numbers overstate protection, and the evidence channel is exploitable. The responsible reading is to keep contestable decisions under human control, gate the evidence channel by item class, and treat the item classifier as a security boundary.

References

- Bai, Y.; Kadavath, S.; Kundu, S.; Askell, A.; et al. 2022. Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*.
- Binns, R. 2018. Fairness in Machine Learning: Lessons from Political Philosophy. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (FAT)*, volume 81 of *Proceedings of Machine Learning Research (PMLR)*, 149–159.
- Dahl, R. A. 1989. *Democracy and Its Critics*. New Haven, CT: Yale University Press.
- Durmus, E.; Nguyen, K.; Liao, T. I.; Schiefer, N.; Askell, A.; Bakhtin, A.; Chen, C.; Hatfield-Dodds, Z.; et al. 2023. Towards Measuring the Representation of Subjective Global Opinions in Language Models. *arXiv preprint arXiv:2306.16388*.
- Haerpfer, C.; Inglehart, R.; Moreno, A.; Welzel, C.; Kizilova, K.; Diez-Medrano, J.; Lagos, M.; Norris, P.; Ponarin, E.; and Puranen, B. 2022. World Values Survey: Round Seven – Country-Pooled Datafile Version 5.0. Madrid and Vienna: JD Systems Institute and WWSA Secretariat. DOI: 10.14281/18241.20.
- Hewitt, L.; Kroner Dale, M.; and de Font-Reaulx, P. 2026. DeliberationBench: A Normative Benchmark for the Influ-

ence of Large Language Models on Users’ Views. *arXiv preprint arXiv:2603.10018*.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.

Inglehart, R.; and Welzel, C. 2005. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press.

Inglehart, R. F. 2018. *Cultural Evolution: People’s Motivations Are Changing, and Reshaping the World*. Cambridge: Cambridge University Press.

Khetan, R.; and Khetan, A. 2026. PoliticsBench: Benchmarking Political Values in Large Language Models with Multi-Turn Roleplay. In *ICML 2026 Workshop on Trustworthy AI for Good*. ArXiv:2603.23841.

Koniaris, M.; Tsipi, A.; and Tsanakas, P. 2026. ParliaBench: An Evaluation and Benchmarking Framework for LLM-Generated Parliamentary Speech. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference (LREC 2026)*, 4797–4818. European Language Resources Association (ELRA). ArXiv:2511.08247.

National Centre for Social Research. 2026. British Social Attitudes Survey, 2022, 2023 and 2024. UK Data Service. SN 9283 (2022), 9363 (2023), 9478 (2024); 2nd editions. DOIs: 10.5255/UKDA-SN-9283-2, 10.5255/UKDA-SN-9363-2, 10.5255/UKDA-SN-9478-2.

Page, B. I.; and Shapiro, R. Y. 1983. Effects of Public Opinion on Policy. *American Political Science Review*, 77(1): 175–190.

Pitkin, H. F. 1967. *The Concept of Representation*. Berkeley, CA: University of California Press.

Santurkar, S.; Durmus, E.; Ladhak, F.; Lee, C.; Liang, P.; and Hashimoto, T. 2023. Whose Opinions Do Language Models Reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML)*. ArXiv:2303.17548.

Sharma, M.; Tong, M.; Korbak, T.; Duvenaud, D.; et al. 2023. Towards Understanding Sycophancy in Language Models. *arXiv preprint arXiv:2310.13548*.

Turner, A. M.; Thiergart, L.; Leech, G.; Udell, D.; Vazquez, J. J.; Mini, U.; and MacDiarmid, M. 2023. Activation Addition: Steering Language Models Without Optimization. *arXiv preprint arXiv:2308.10248*.

Zou, A.; Phan, L.; Chen, S.; Campbell, J.; Guo, P.; Ren, R.; et al. 2023. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint arXiv:2310.01405*.